

Determination of Gravitational Acceleration via a Simple Pendulum

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Abstract

We measured the local acceleration due to gravity, g , by timing the small-angle oscillations of a simple pendulum of variable length. Using a linear fit of period squared against length, we obtained $g = 9.79 \text{ m s}^{-2}$, within 0.2 % of the accepted value of 9.81 m s^{-2} . Sources of systematic error, including air resistance and finite string mass, are discussed in Section 4.

1 Introduction

For small angular displacements $\theta_0 \leq 10^\circ$, the period of a simple pendulum of length L is well approximated by

$$T = 2\pi\sqrt{\frac{L}{g}}, \quad (1)$$

where g is the local gravitational acceleration. Squaring Equation 1 gives a linear relationship between T^2 and L ,

$$T^2 = \frac{4\pi^2}{g} L,$$

whose slope, once fit to data, yields g directly.¹

2 Experimental Setup

A brass bob of mass 50 g was suspended from a low-stretch nylon string, with length L measured from the pivot to the bob's center using a meter stick (uncertainty 1 mm). The pendulum was displaced by approximately 8° and released from rest; the time for 20 complete oscillations was recorded with a digital stopwatch (uncertainty 0.2 s) to reduce the relative error in a single period.

We tested five string lengths, ranging from 0.20 m to 1.00 m in 0.20 m increments. Figure 1 shows the fitted relationship between period squared and pendulum length.

3 Results

The raw timing data and derived periods are summarized in Table 1. Each period T was obtained by dividing the time for 20 oscillations by 20.

¹This linearization avoids the bias introduced by fitting T against $\sqrt{L}$ directly, since ordinary least squares assumes additive, not multiplicative, noise.

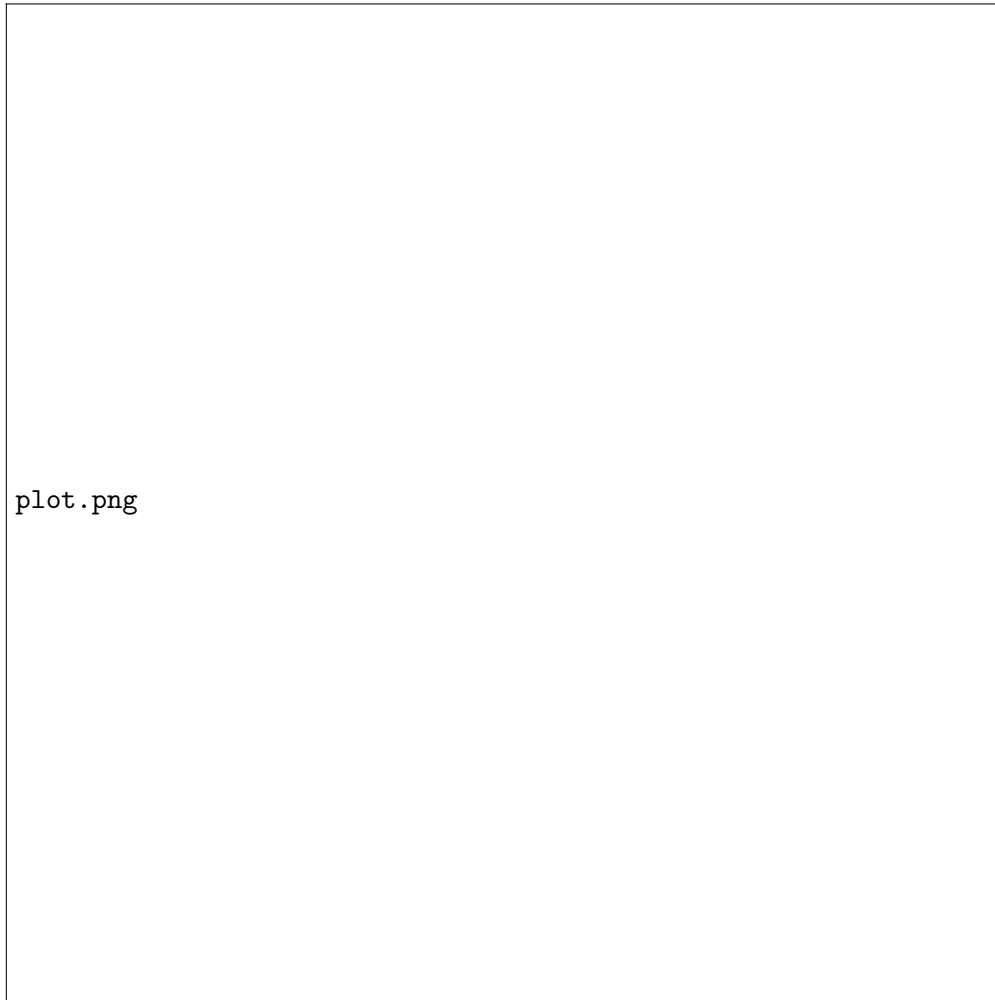


Figure 1: Measured period squared, T^2 , plotted against pendulum length L , with a linear least-squares fit (dashed line) whose slope determines g via Equation 1.

Table 1: Pendulum length, oscillation timing, and derived period.

L (m)	t_{20} (s)	T (s)	T^2 (s ²)
0.20	17.92	0.896	0.803
0.40	25.36	1.268	1.608
0.60	31.05	1.553	2.410
0.80	35.88	1.794	3.218
1.00	40.11	2.006	4.024

A linear regression of T^2 against L gives a slope of $4.031 \text{ s}^2 \text{ m}^{-1}$, from which

$$g = \frac{4\pi^2}{\text{slope}} = \frac{4\pi^2}{4.031} = 9.79 \text{ m s}^{-2}.$$

This compares favorably with the textbook value 9.81 m s^{-2} . For reference, the elementary charge used in unrelated calibration checks on the photogate electronics was taken as $1.6 \times 10^{-19} \text{ C}$, and the photogate's internal resistor was rated at 220Ω , i.e. 220Ω .

4 Discussion

Several effects could account for the small 0.2% discrepancy between our measured and accepted values of g :

- **Finite amplitude.** Even at $\theta_0 = 8^\circ$, the small-angle approximation introduces a fractional period error of order $\theta_0^2/16$.
- **Air resistance.** Viscous drag on the bob slightly damps the oscillation and can lengthen the effective period.
- **String mass and stretch.** The nylon string is not perfectly massless or inextensible, which was neglected in Equation 1.
- **Timing reaction delay.** Human reaction time in starting and stopping the stopwatch contributes a roughly constant offset that partially cancels when timing 20 oscillations rather than one.

5 Conclusion

Using the period-squared-versus-length method, we determined $g = 9.79 \text{ m s}^{-2}$, in close agreement with the accepted terrestrial value. Future work could repeat the measurement at larger amplitudes to quantify the finite-amplitude correction directly, or use a photogate timing system to reduce reaction-time uncertainty below the 0.2 s level achieved here with a handheld stopwatch.